Worksheet 5.3

A world of reactions

Answers

<table>
<tr><th>Balancing a chemical equation</th><th>Example: Methane gas will burn in air. This is a combustion reaction. This type of reaction produces CO_2 and H_2O.</th></tr>
<tr><td>Step 1: Start with the word equation and name all of the reactants and products.</td><td>methane gas + oxygen gas $\longrightarrow$ carbon dioxide + water</td></tr>
<tr><td>Step 2: Replace the words in the word equation with formulae and rewrite the equation.</td><td>Methane gas = CH_4 Oxygen gas = O_2 (reactants)
Carbon dioxide = CO_2 Water vapour = H_2O (products)
$CH_4 + O_2 \longrightarrow CO_2 + H_2O$</td></tr>
<tr><td>Step 3: Count the number of atoms of each element (represented by the formulae of the reactants and products).</td><td><table><tr><th>Element</th><th>Reactants</th><th>Products</th></tr><tr><td>C</td><td>1</td><td>1</td></tr><tr><td>H</td><td>4</td><td>2</td></tr><tr><td>O</td><td>2</td><td>3</td></tr></table></td></tr>
<tr><td>Step 4: If the number of atoms of each element is the same on both sides of the equation, the equation is already balanced. If not, numbers need to be placed in front of one or more of the formulae to balance the equation. These numbers are called coefficients and they multiply all of the atoms in the formula.</td><td>To balance the hydrogen atoms, put a 2 in front of H_2O:
$CH_4 + O_2 \longrightarrow CO_2 + 2H_2O$.
The oxygen atoms can be balanced by putting a 2 in front of the O_2 on the left:
$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$.
The equation is now balanced. It can be checked by counting the number of atoms of each element on both sides of the new equation.<table><tr><th>Element</th><th>Reactants</th><th>Products</th></tr><tr><td>C</td><td>1</td><td>1</td></tr><tr><td>H</td><td>4</td><td>4</td></tr><tr><td>O</td><td>4</td><td>4</td></tr></table></td></tr>
<tr><td>Step 5: Add physical state symbols.</td><td>$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$</td></tr>
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<tr><td>Step 2: Replace the words in the word equation with formulae and rewrite the equation.</td><td>Ethene gas = C_2H_4 Oxygen gas = O_2 (reactants)
Carbon dioxide = CO_2 Water vapour = H_2O (products)
$C_2H_4 + O_2 \longrightarrow CO_2 + H_2O$</td></tr>
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